

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is

The logs show that: Starting at packet 52 (about 3.39s), IP 203.0.113.0 sends a continuous stream of TCP SYN packets to the web server at 192.0.2.1, all from port 54770 to 443. The same address sent about 140 SYN packets and never completed a handshake. Normal visitors (198.51.100.x) completed handshakes and got 200 OK early on. After the flood began, their connections failed with RST, ACK resets, and a 504 Gateway Time-out.

This event could be : A TCP SYN flood attack, which is a type of denial of service (DoS) attack. The traffic comes from a single IP address (203.0.113.0), so it is DoS rather than DDoS.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The client sends a SYN packet to the server to request a connection.
2. The server replies with a SYN-ACK packet to acknowledge the request and reserve resources for the connection.
3. The client sends a final ACK packet to confirm, and the connection is established.

Explain what happens when a malicious actor sends a large number of SYN packets all at once: The server replies to each SYN with a SYN-ACK and waits for the final ACK, which never arrives. These half-open connections fill the connection queue and use up the server's resources, so it can't accept new connections

Explain what the logs indicate and how that affects the server: The logs show 203.0.113.0 sending a continuous flood of SYN packets with no completing ACKs. Legitimate visitors start receiving RST, ACK resets, and a 504 Gateway Time-out. The server is overwhelmed, and the website becomes unavailable, causing connection timeout errors.

